

PaperAgent

OPEA-based Cloud-Edge Academic Intelligence

AI for Good OPEA Competition Technical Report

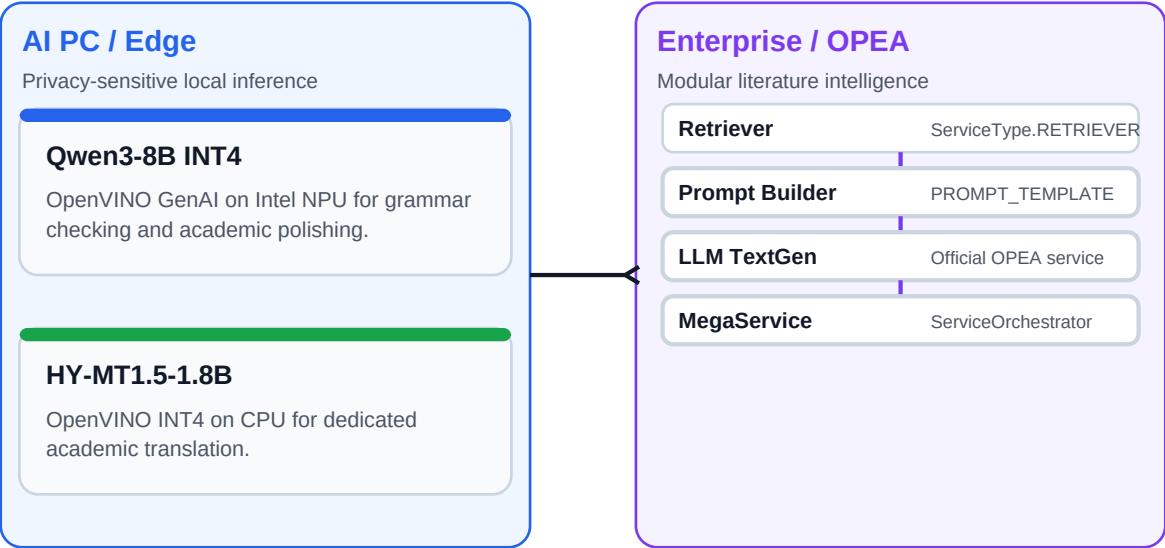
Problem

Cloud-only academic AI can expose sensitive drafts, while local-only models are not ideal for scalable literature retrieval and enterprise knowledge orchestration.

Solution

PaperAgent separates privacy-sensitive writing tasks onto the AI PC and composes literature retrieval and grounded generation as modular OPEA services.

Architecture Overview



AI for Good / Enterprise Value

PaperAgent lowers the privacy and infrastructure barriers of AI-assisted academic work. Sensitive drafts remain on the user device, while OPEA provides a reusable cloud-native path for retrieval, prompt construction, model integration, and grounded knowledge assistance.

Local privacy

Real model inference

OPEA composition

Open-source delivery

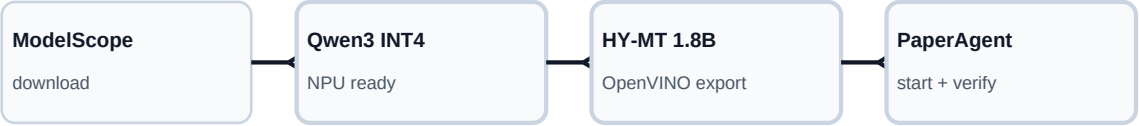
Technical Implementation & Reproducibility

OPEA Component Mapping

Layer	Implementation	OPEA Role
Retrieval	retriever_service.py	RETRIEVER MicroService
Prompt	prompt_service.py	PROMPT_TEMPLATE MicroService
Generation	opea/llm-textgen	LLM MicroService
Orchestration	megaservice.py	ServiceOrchestrator + MegaService

One-click Model Preparation

deploy.bat prepares the real runtime models before starting the application. ModelScope is the default source; Hugging Face remains an automatic fallback when needed. Model weights are not committed to Git.



Deployment

- 1 Run .\deploy.bat
- 2 Models prepared and verified
- 3 Edge runtime + UI prepared
- 4 OPEA Docker stack deployed
- 5 Sensitive-data scan + health checks

Evaluation Focus

Review the complete prototype: local grammar/polish on Qwen3 NPU, local HY-MT translation on CPU, OPEA literature RAG, topology verification, sanitized demo data, automated deployment, and tracked-source sensitive-data scanning. The project avoids unsupported performance claims and focuses on functional integration, privacy-aware workload placement, modularity, and reproducibility.

Open-source commitment

PaperAgent source code is released under Apache License 2.0. Third-party frameworks and models retain their upstream licenses. Credentials, private corpora, runtime uploads, model weights, and internal network information are excluded from the competition repository.

Demo guide: docs/demo-guide.md | Architecture: docs/*.svg | Report source: technical-report.md